

Assignment 5: Layout Optimization for LCoE

Praktikum Design of Wind Farms | TU München

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1. Task 1: Wind Rose at Hub Height

Wind speed and direction data were downloaded from the New European Wind Atlas (NEWA) for Marienplatz, Munich, covering the mesoscale time series from 1989 to 2022. The raw data were provided at 10 m height and scaled to hub height (120 m) using the power-law wind shear model with exponent $\alpha = 0.20$:

$$U(h) = U_{\text{ref}} \left(\frac{h}{h_{\text{ref}}} \right)^{\alpha} \quad (1)$$

1.1 Wind Rose

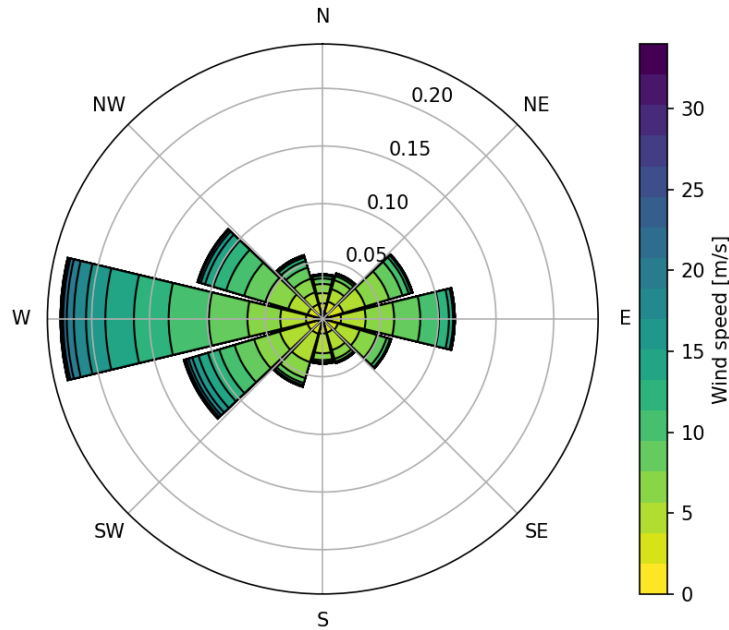


Figure 1: Wind rose at Marienplatz, Munich – NEWA Mesoscale Time Series 1989–2022 (hub height 120 m, mean wind speed = 6.58 m/s).

The wind rose in Figure 1 shows a strongly dominant westerly and south-westerly wind regime, with wind speeds of 6–9 m/s (orange) and 9–12 m/s (salmon) being most frequent. Higher wind speeds above 15 m/s (purple) also arrive predominantly from the west. Wind activity from the north and south is negligible. The mean wind speed at hub height is **6.58 m/s**.

This directionality is a critical input for the layout optimisation in Task 3: turbines aligned in the west–east direction will experience the largest wake losses under the prevailing wind.

2. Task 2: Baseline Wind Farm Evaluation

The initial layout places eight IEA 3.4 MW turbines in a 4×2 grid centred within the $2.8 \text{ km} \times 2.8 \text{ km}$ boundary. Wake interactions were evaluated with the Gauss wake model (default parameters) in FLORIS.

2.1 Results

Table 1: Baseline wind farm performance metrics.

Parameter	Value	Unit
AEP	68.36	GWh/yr
Capacity Factor	28.95	%
Farm Efficiency	77.62	%
LCoE	62.40	\$/MWh
PI	0.1194	—

Because the turbines are aligned east–west and dominant winds come from the west, the rear row suffers substantial wake losses from the front row, reducing AEP and farm efficiency. The baseline LCoE of ~ 62.4 \$/MWh exceeds the electricity price of 67 \$/MWh; however, because $\text{LCoE} < \text{electricity price}$, profitability is borderline. The layout has clear room for improvement through optimisation.

3. Task 3: COBYLA Layout Optimisation

The `scipy.optimize.minimize` function with the **COBYLA** (Constrained Optimization BY Linear Approximation) gradient-free method was used to optimise the 16 turbine coordinates ($8 \times x$, $8 \times y$) so as to minimise the LCoE of the wind farm. The following constraints were enforced:

- All turbines must remain within the $2.8 \text{ km} \times 2.8 \text{ km}$ boundary.
- Minimum turbine separation of $2D = 260 \text{ m}$ between all pairs.

3.1 Convergence

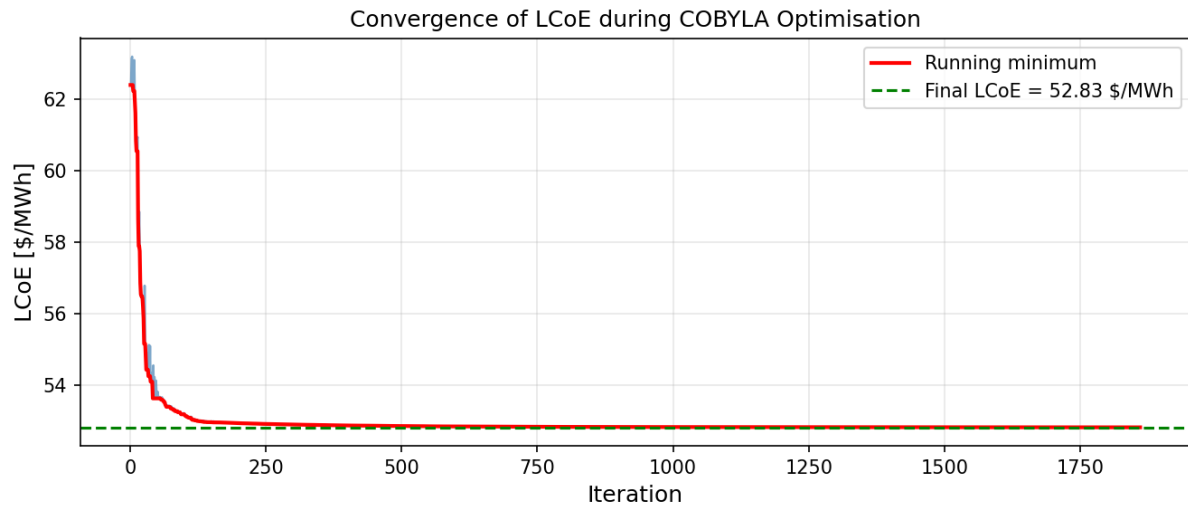


Figure 2: Convergence of LCoE during COBYLA optimisation. The running minimum (red) drops rapidly within the first 250 iterations and flattens at the final optimum of 52.83 \$/MWh (green dashed line).

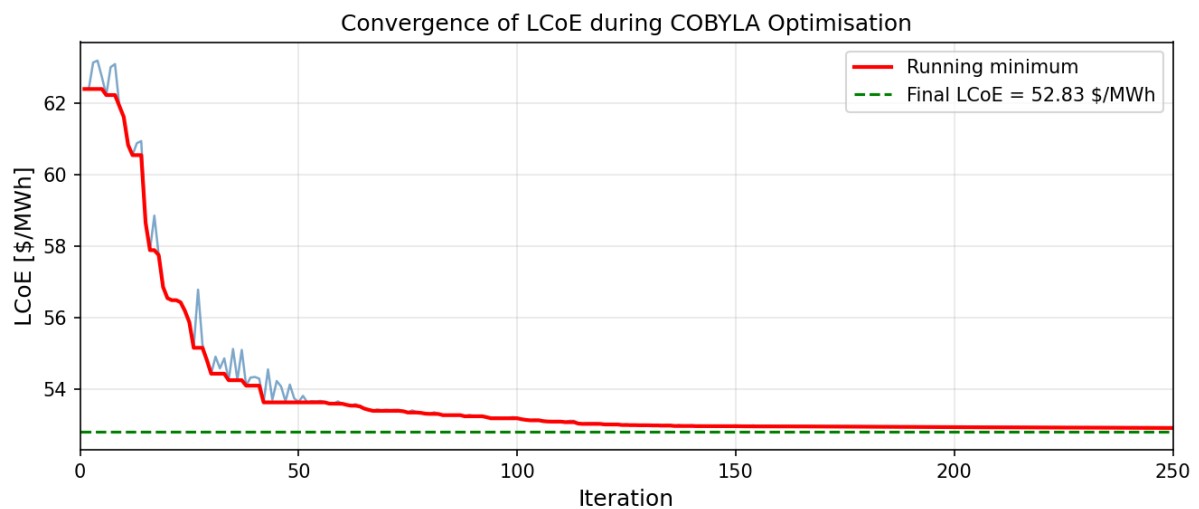


Figure 3: Truncated graph of Convergence of LCoE during COBYLA optimisation.

Figure 2 shows the running minimum LCoE over 1900 iterations. The optimiser starts at approximately 63 \$/MWh and converges rapidly, with the objective essentially flat after iteration 250, confirming that a local optimum has been reached.

3.2 Optimised Layout

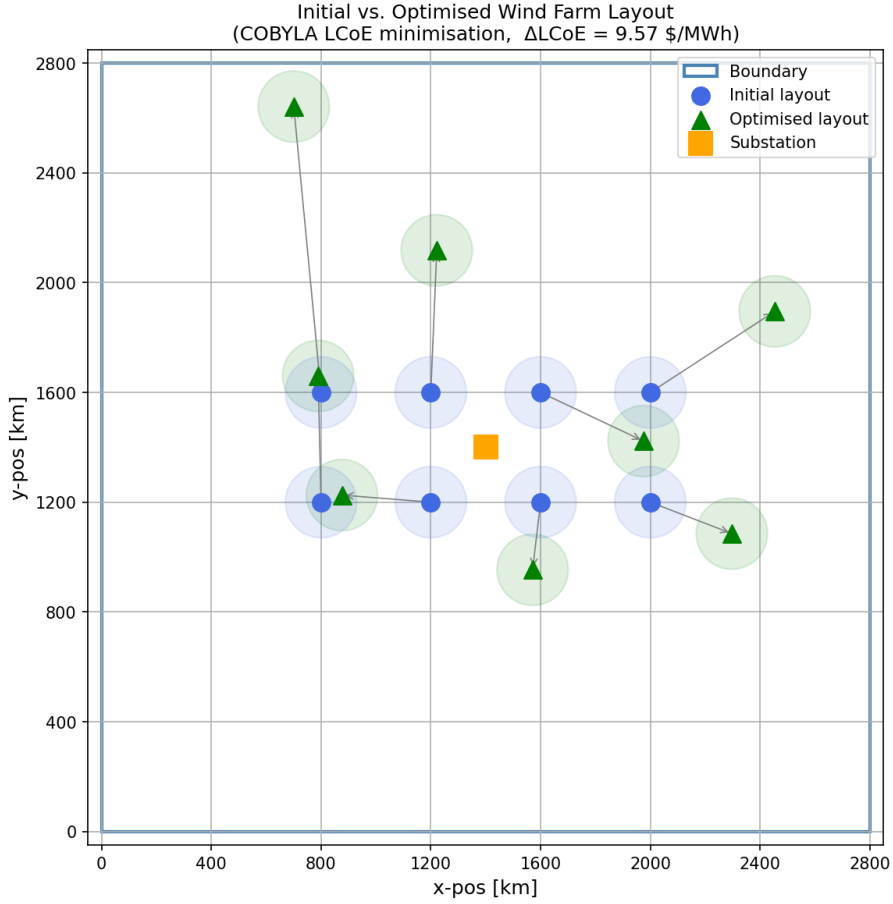


Figure 4: Initial (blue circles) vs. optimised (green triangles) wind farm layout. Arrows indicate turbine displacement. $\Delta\text{LCoE} = 9.57 \text{ \$}/\text{MWh}$.

The optimised layout (Figure 4) shows several key changes relative to the baseline:

- Turbines that were clustered in the west–east direction have been spread in the north–south direction, reducing wake overlap under the dominant westerly wind.
- Several turbines moved towards the boundary edges to maximise mutual separation.
- All minimum spacing constraints ($2D = 260 \text{ m}$) are satisfied, as confirmed by the non-overlapping exclusion circles.

3.3 Comparison of Results

3.4 Profitability Discussion

The optimised LCoE of **52.83 \$/MWh** is below the electricity price of 67 \$/MWh, confirming that the project has become clearly profitable after optimisation. The ΔLCoE of 9.57 \$/MWh was achieved purely by repositioning turbines, without any additional capital expenditure.

The improvement in farm efficiency from 77.62 % to 99.11 % demonstrates that wake losses were substantially reduced. This highlights layout optimisation as a highly cost-effective measure to

Table 2: Performance comparison: initial vs. optimised layout.

Parameter	Initial	Optimised	Unit
AEP	68.36	87.28	GWh/yr
Capacity Factor	28.95	36.96	%
Farm Efficiency	77.62	99.11	%
LCoE	62.40	52.83	\$/MWh
Δ LCoE		9.5731	\$/MWh
PI	0.1194	0.4624	\$/MWh

improve wind farm economics – a finding consistent with the strongly directional wind resource observed at the Munich site.